

Simulation-Based Inference

Organization and Overview

Paul Bürkner

Simulation-Based Inference (SBI)

- Lecture and exercises
- Both theoretical and practical aspects
- For application in all quantitative sciences
- Help you solve real-world problems

Lectures:

- Tuesdays 14:15-15:45, Room: M/E25 in the Math Tower
- Thursdays 10:15-11:45, Room: M/E25 in the Math Tower

Tutorials to discuss the exercises:

- Time: Mondays 12:15-13:45
- Room: M/E21 in the Math Tower

At the end of this course, you should know...

- a lot of important concepts in and around simulations
- how simulations benefit statistical workflows
- how to use simulations for inference
- how to use the BayesFlow library
- how to write reports about data analysis projects
- how to apply all of this in real-world scenarios

Professor for Computational Statistics



Selection of research interests

- Uncertainty quantification
- Prior specification
- Model comparison
- Simulation-based inference
- Amortized Bayesian inference
- Machine-assisted workflows

Search for

- Simulation-Based Inference (SoSe26)

Will contain all the course-related materials

- Lecture slides (uploaded after each lecture)
- Exercises and solutions

- Put on Moodle every week on Thursdays

Tutorials discussing the exercises:

- Time: Mondays 12:15-13:45
- Room: M/E 21 in the Math Tower
- Lead by members of my group
- Not mandatory and not graded
- Highly recommended for deeper understanding
- First tutorial: 27.04.25

- The Moodle course has a public forum
- Please use it to ask public questions
- Feel free to engage in others' questions as well!

Some knowledge about

- Probability theory
- Statistics or machine learning
- Linear algebra and calculus
- High-level programming languages (e.g., R or Python)

Who are you actually?

Interact with us during the course!

- Ask questions at any point
- Tell us if something remains unclear
- Give input on what you would like to learn

- If you are studying in the Master Data Science, you can choose if you want to do it in the “application” or “methods” module
- If you are studying in the Bachelor Data Science, you have to do it in the “application” module

- No classical exam!

Project work:

- A real-world data analysis using SBI
- In groups of 1-3 people
- A single report per group (graded)
- A single short presentation per group (graded)
- Details and criteria will be discussed in time

Major topics of this course

- Reasons for using simulations
- Pseudo-randomness with computers
- Sampling from simple distributions
- Simulation-based tests
- Resampling methods
- Frequentist simulation studies
- Introduction to Bayes
- Likelihood-based Bayesian inference
- Simulation-based verification of Bayesian inference
- Simulation-based prior elicitation
- Approximation Bayesian computation
- Neural networks and generative neural models
- Neural Bayesian Inference

Any remaining questions about the organization?